

Heterogeneous Multi-Robot Person-Following in Constrained Environments

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Abstract—There are many reasons for a robot to track a moving person, from security monitoring to providing social assistance. In this work, we consider the problem of autonomously tracking a moving person through an altitude-constrained environment using mobile robotic platforms. This problem presents several challenges. Because we are constrained to low altitudes, such as in indoor environments or below the roof or treeline in an outdoor environment, in order to maintain sensor tracking the platforms must strongly consider obstacles and occlusion. Because we consider the robot's maximum speed to be constrained to the same order of magnitude as the target, the robots must make timely and correct navigation decisions to maintain track. We examine solutions to these challenges using a robot team consisting of a slower, more-capable ground robot and a faster, less-capable micro aerial vehicle. Three methods for robot navigation decision-making are evaluated: a heuristic-based predictive goal search algorithm, a multi-agent reinforcement learning method, and a baseline closed-loop method. While the heuristic method proved the most reliable, completing all nine paths and reaching targets nearly twice as fast as RL, we were still able to demonstrate a successful MARL-based solution. These findings reveal both the potential and limitations of RL, demonstrating the promise of heterogeneous multi-robot systems for person-following in unpredictable settings.

I. INTRODUCTION

For robots to be useful in various settings, they will need to interact with other agents in shared environments. This increase in interactions adds complexity to navigation and operation, requiring robots to perform tasks efficiently and adapt to human behaviors [1]. In more complex environments, heterogeneous systems that incorporate different types of robots can improve task outcomes [2].

In this work, we utilize a heterogeneous team of robots to address the problem of person following in challenging environments. Specifically, we pair a faster aerial robot (UAV) with a slower ground robot (UGV) with more sensing and computation capability, where the UAV is better able to keep up with the target and the UGV is able to help task and localize the aerial robot. We consider non-cooperative target movement and constraints such as navigation space (altitude), sensing, and speed limitations. We compare strategies for coordinating these robots despite their distinct capabilities and limitations.

First, we introduce a method that leverages predictive goal-searching algorithms to ensure robust person-following

in height-restricted environments, Forecasting Optimization of Robotic Way-point Assignment for Robust and Dynamic Person Following (FORWARD-PF). In the FORWARD-PF method, the UGV's objective is to navigate to the target's position with the UAV assisting. The UGV uses its perception to construct a motion model, anticipating the target's future movements. This model helps the UGV follow or approach the target. Meanwhile, the UAV aims to find optimal positions in real-time for maximum visibility of the target. For example, if the target moves around a corner, the UAV should position itself to maintain continuous visibility. We also introduce a multi-agent reinforcement learning (MARL) solution to the problem, and conduct experiments to compare these two new methods against a baseline: FORWARD-PF, which systematically searches the environment and assigns the highest-scoring locations to each robot; our multi-agent reinforcement learning (MARL) approach, where agents learn optimal strategies through interaction and rewards; and a baseline UGV-only approach, where the UGV moves directly towards the target.



Fig. 1. Unity-based simulation environment containing an Clearpath Robotics Warthog [3], circled in red, a generic quad-rotor aerial robot, circled in blue, and a target (e.g., soldier model) circled in yellow.

II. RELATED WORK

1) *Person-Following*: Numerous person-following solutions exist, tailored to specific objectives, contexts, and team dynamics, with no single approach universally optimal. These include single ground robots, single aerial robots, social ground robots, and heterogeneous multi-robot teams.

According to [4], person-following approaches can be classified into three main categories: position-based, image-based, and end-to-end learning-based. This work looks at two of these with the FORWARD-PF and RL-Person method falling into the position-based, and the closed-loop baseline being an image-based approach.

We also focus on the challenging space of air-ground robot teams performing person following with specific constraints

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such as height, speed (walking at a reasonable speed), avoiding obstacles, taking smooth turns, and handling complete occlusion using an aerial robot to maintain visibility, without any prior knowledge about the target other than maximum velocity bounds, which has not been previously well-addressed.

2) *Visibility-Based Pursuit-Evasion*: The visibility-based pursuit-evasion problem defined by [5] involves maintaining line-of-sight with evaders in a 2D environment with obstacles, critical for tasks like surveillance and search and rescue. Traditional methods prioritize visibility over speed, focusing on navigating around obstacles to keep visual contact. However, these methods are less effective for tasks requiring close following, as their goal is only line-of-sight and they assume evasive target behavior, unlike the non-cooperative targets in this work. Despite these differences, the problem informs the aerial robot's role in our system.

3) *Heterogeneous Multi-robot Teaming*: [6] used human gaze to guide UAVs, highlighting challenges in coordinating heterogeneous robots and human interactions. Another study [7] combined aerial and ground robots for 3D surveillance, leveraging their complementary strengths. These approaches inform this research, which addresses the limitations of single robot systems by developing strategies for a heterogeneous team to enhance person-following in shared environments.

4) *Multi-agent Reinforcement Learning*: This paper utilizes a MARL method for pursuit-evasion tactics inspired by [8]. We use Multi-Agent MADDPG and PettingZoo's MPE for training in simulation, although this work involves the challenge of generalizing deployment onto a non-omnidirectional platform. Challenges in simulation-to-real transfer include the simulation-reality gap and discrepancies leading to failures [9]; this proved a challenge during the course of this work.

III. PROBLEM DEFINITION

One Differential Drive Robot (DDR), one holonomic robot, and one holonomic agent are modeled as oriented discs in an unexplored 3D environment with obstacles. The DDR is the follower, the holonomic robot is the observer, and the holonomic agent is the target. The follower and observer have sensors to detect the target if an unobstructed line of sight exists. The target moves independently toward its destination, with a greater maximum velocity than the follower but less than the observer. The observer maintains visibility of the target. This work aims to develop motion strategies for the follower and observer to achieve their goals. The strategies are based on the target's current or last known position.

IV. METHODS

A. FORWARD-PF

The FORWARD-PF approach systematically searches the environment and scores candidate locations based on the objective of each robot. The locations with the highest scores are then assigned to the respective robots (Algorithm 1).

Algorithm 1 Forecasting Optimization of Robotic Way-point Assignment for Robust and Dynamic Person Following (FORWARD-PF)

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1: Input: Time-horizon  $z$ , Observer optimal distance from target  $d_s$ 
2: while do
3:   Initialize candidate goal locations  $C$  around current position of target
4:   Generate motion model  $P(t)$  using past position observations of the target
5:   Generate motion prediction points  $F_p(z)$  using  $P(t)$  over time-horizon  $z$ 
6:   for each candidate goal location  $c$  in  $C$  do
7:     Score  $c$  using  $f\_score \rightarrow$  distance between  $c$  and average of  $F_p(z)$ 
8:     Score  $c$  using  $o\_score \rightarrow$  number of points in  $F_p(z)$  with line-of-sight to  $c$  minus  $d_s$  minus distance between  $c$  and average of  $F_p(z)$ 
9:   end for
10:  Select  $c_{f\_best}$  with the lowest  $f\_score$  in  $C$ 
11:  Select  $c_{o\_best}$  with the highest  $o\_score$  in  $C$ 
12:  Score current robot position
13:  if Current robot position is better than  $c_{f\_best}$  or  $c_{o\_best}$  respectively then
14:    Don't move the robot
15:  else
16:    Task  $c_{f\_best}$  to follower and  $c_{o\_best}$  to observer
17:  end if
18: end while

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In this formulation, time-horizon refers to the number of seconds a motion model of a target is predicted into the future. Position history refers to the number of seconds in the past over which position data is collected of a target and used to calculate a motion model. Motion prediction refers to the process of estimating the future motion or trajectory of an object or agent based on its current state and past observations.

The goal of motion prediction is to anticipate the future behavior of the target in order to make informed decisions. This approach uses a third degree polynomial to model the target's motion, $P(t)$. Using position data over the position history, a polynomial fitting algorithm is used to calculate the coefficients. The input is the time in seconds the position was observed and the output is a set of coordinates, (x, y). Points are then generated with this polynomial by getting coordinates between current time and the time-horizon in one second intervals. Each point is then used as the input for a normal Gaussian distribution with a σ of 0.9 generating 10 points each. These points, F_p , are meant to characterize the potential motion of the target over the time-horizon.

The set of candidate goal locations, C , is generated as a grid around the target's current position. Prior to the experiment execution, two values need to be defined. The value n which defines the size of the grid as the order value

of a square matrix, meaning the grid would be of size $n \times n$, we will refer to this as C_{rows} . The other value is the distance in meters that each cell of the grid is spaced out by, $C_{spacing}$. For example, if $C_{spacing} = 1.0$, for all cells c in C , the center of cell c is 1 meter distance from all of its neighboring cells. From these two values, our set of candidate goal locations are defined. These values should be tweaked based on the environment in which this system is running. An obstacle-free outdoor space can afford a higher values for both since staying close to the target isn't as necessary when compared to an indoor environment with lots of hallways where the threshold for an optimal goal location could be small. Once the set of candidate goal locations are generated, scoring begins.

There are two goal location scoring formulas, one for each agent of the pursuing team. Eq. 1, where goal locations are scored with the objective of the follower to achieve and maintain a close proximity distance to the target. Similarly, Eq. 2, where goal locations with the objective of the observer maintaining view of the target over some time-horizon, while also minimizing travel time to a new goal location.

Each candidate goal location is scored for the follower by the distance between the goal location and the value of the predictive motion polynomial at the time-horizon z :

$$follower_{score} \rightarrow ||c - avg(F_p(z))||_2 \quad (1)$$

Each candidate goal location's score is computed for the observer as utility minus cost. The utility is the line-of-sight score which is the number of predicted motion points, F_p , visible to candidate goal location c . Function $los(c, F_p)$ returns the number of predicted motion points f in F_p that are visible to candidate goal location c in C . Costs includes the straight-line distance from the robot platform to the person, and the difference in distance between the preferred distance around the person to maintain, d_s , and the distance the candidate goal location is from the person.:

$$observer_{score} \rightarrow los(c, F_p) - |d_s - ||c - avg(F_p(z))||_2| \quad (2)$$

This process operates iteratively. When the target is in motion, the robots are continuously assigned goal locations that align with their objectives. Upon the target reaching its final destination and ceasing movement, the robots should maintain their goal positions. Given the target's stationary status and the unchanging environment, the evaluation of potential goal locations will yield consistent scores at each time step.

B. RL-Person

The Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm is employed for this multi-agent reinforcement learning (MARL) approach. Utilizing a centralized critic in a Centralized-Training Decentralized-Execution (CTDE) framework, MADDPG addresses non-stationarity and supports both cooperative and competitive scenarios [8].

In our setup, three agents, Target, Follower, and Observer are trained simultaneously. The Target evades pursuers, the Follower aims to stay close to the Target, and the Observer

maintains visual contact while avoiding obstacles. Each agent provides its position, velocity, and relative positions and velocities of other agents and obstacles as observations. The action space includes movements in four directions or no action. The trained model is deployed on the robot system, using real-time observations to compute actions.

C. Closed-loop Baseline

The closed-loop method works by maintaining the followers heading directed towards the target. This is a single robot method designed to stand as a baseline motion strategy against the previous two which each utilize two robots on the pursuer team.

This method thus allows for the robot to adjust its heading and maintain a steady approach towards the target by moving forward while consistently re-centering the follower so it is always heading towards the target.

V. EXPERIMENTS

A. Experiment Design

The experiment, conducted in a Unity-based ROS simulator, begins with the ground and aerial robots spawning near each other, while the target starts out of sight. The target follows a predefined path, and the experiment officially begins when the target enters the robots' view. The experiment evaluates the multi-agent system's ability to follow a target across paths of increasing complexity (Fig. 2).

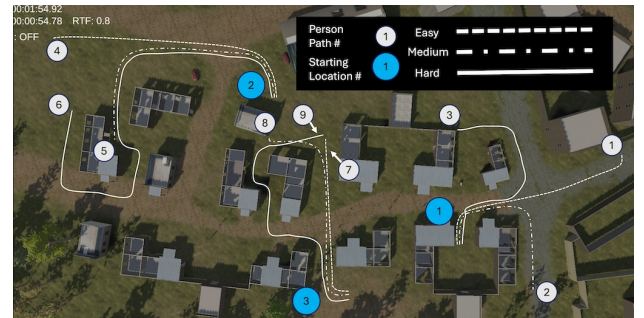


Fig. 2. Top-down view of the Unity environment where the experiments take place. The numbered blue circles show the experiment starting position for the neighboring paths. The numbered white circles indicate the end of a path and its number. The paths pattern illustrates the tortuosity/difficulty of the path.

B. Metrics

The following metrics evaluate each method's success, efficiency, and robustness:

Success Rate: The percentage of experiments where the UGV successfully navigates to the target's stationary end position. Failure occurs if the robot collides with an object or loses visual contact with the target in a way that prevents reacquisition. There is no time limit, but experiments are terminated and marked as failures if the experimenter deems reacquisition infeasible.

Time in View: This metric, inspired by [10], measures the percentage of time the target is visible to the robot

system. It reflects the UAV's effectiveness in maintaining target visibility, which helps the UGV navigate.

Cumulative Distance: The total distance traveled by the UGV from start to the target's end position. A lower value suggests more efficient navigation and decision-making.

Maximum Distance: The farthest distance the UGV reaches from the target during the experiment. A higher value may indicate less efficient navigation. Point-and-shoot methods may struggle with visual contact, while FORWARD-PF and MARL approaches are expected to perform better by adapting to the environment and coordinating between robots.

C. Closed-loop Baseline

In our simulation, the closed-loop method serves as a baseline. It uses a forward-facing camera to detect the target and calculate the angular velocity needed to keep the target centered in the camera frame. This method involves adjusting the robot's heading to keep the target's bounding box centered, while maintaining a fixed forward velocity. The robot uses real-time visual feedback to follow the target across all nine paths.

D. FORWARD-PF

For this method, robots use a full autonomy stack for mapping and path planning. The simulated camera identifies the target, and motion prediction integrates target position and velocity using an extended Kalman filter to reduce sensor noise. This data is used to project the target's future position with a third-degree polynomial and a fixed proportion coefficient.

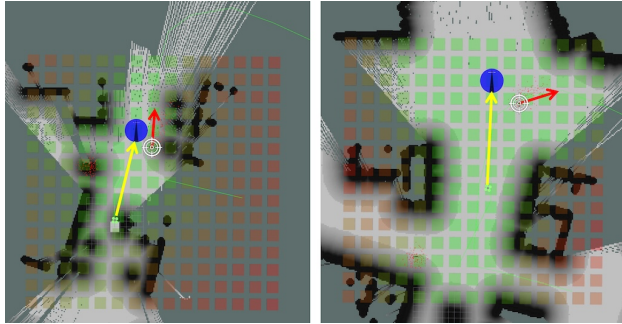


Fig. 3. Snapshots from the heuristic grid-search method showing candidate goal scoring for the UAV. Goal locations are color-coded from green (highest score) to red (lowest score). The blue circle marks the selected goal location, and arrows indicate the UAV's direction and the target's position and motion.

Candidate goal locations are generated in a grid around the target. Each location is scored based on proximity to the target for the ground robot (within a five-meter buffer) and line of sight for the aerial robot. The UAV's score also includes line-of-sight to the predicted motion and obstacle costs from the occupancy grid. The location with the highest score is chosen for navigation. The aerial platform maintains its heading towards the target to ensure continuous camera alignment and accurate tracking.

E. RL-Person

We used PettingZoo, a Python library for multi-agent reinforcement learning (MARL), in conjunction with OpenAI's Gym environments. The Multi Particle Environment (MPE), a 2D space with particles of varying sizes and functions, was used for training MADDPG models (Fig. 4).

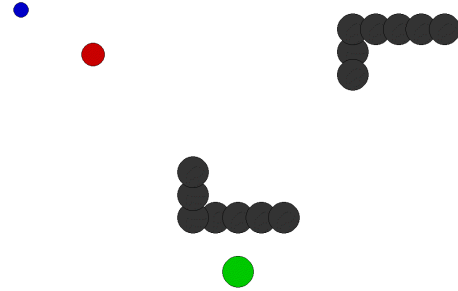


Fig. 4. The Multi Particle Environment (MPE) is used to train MADDPG models in MARL experiments. It includes gray obstacle landmarks, a red target evading the other agents, a faster blue observer agent, and large yet slow green follower agent.

In our MPE, gray particles act as obstacles, the red particle is the evading target, the smaller blue particle is a fast aerial observer, and the larger green particle is a slower ground robot.

Training and Adapting the Model: This experiment trained a follower and observer agent to navigate complex paths, with the follower minimizing distance to the target and the observer maintaining visual contact. The 2D model was adapted for a 3D environment to enable deployment on real robots, transitioning from discrete simulation actions to continuous commands.

Actions were averaged for smooth transitions to continuous motion, and a local planner was used to avoid collisions caused by inconsistently detected obstacles in the 3D environment. The observation space was streamlined by segmenting it into fixed regions, ensuring consistent performance despite varying obstacle counts, with the aerial platform aiming for balanced proximity to the target.

The model's positional data was scaled to adapt from a small training environment to a larger real-world space, improving alignment through fine-tuning. Generalizing the model remains challenging, but adapting it for different environments is currently more feasible than training new models for each case.

F. Real-world Validation

A real-world validation was conducted to demonstrate the practical functionality of the heuristic method on real hardware. Because robust localization was challenging for the UAV platform we used (a Crazyflie), we used an OptiTrack external sensing system within a confined lab space, which restricted configuration of experimentation.

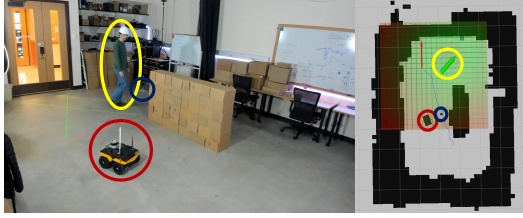


Fig. 5. Our real-world experiment side-by-side with RVIZ containing an autonomous ground vehicle, circled in red, an autonomous micro-aerial vehicle, circled in blue, and a target agent (e.g., a person) circled in yellow.

VI. RESULTS

The simulation experiments evaluated three methods: closed-loop (CL), FORWARD-PF (PF), and RL-Person Following (RL) across varying path complexities: easy, medium, and hard (Fig. 2). The success rates showed that the FORWARD-PF method was the most reliable, successfully completing all paths regardless of complexity. The RL method had a moderate success rate, performing well in simpler paths but failing in more complex ones primarily due to collisions. The CL method only succeeded on the easiest paths. Failures were typically due to collisions or the robot losing sight of the target.

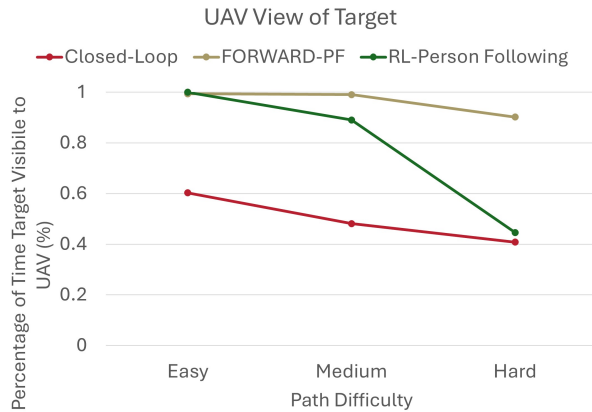


Fig. 6. Percentage of time target is visible to robot team. The performance of the methods are grouped by path difficulty.

The Arrival Time Delta metric, which measures the time difference between the target stopping and the UGV reaching the target, indicated that the PF method outperformed the RL method, arriving nearly twice as fast on average. The CL method demonstrated the smallest Arrival Time Delta on the paths it completed due to its reactive nature that requires no planning time. Metrics such as the cumulative distance traveled and the maximum distance between the UGV and the target also provided insights; the CL method showed the lowest cumulative distance and maximum distance for the configurations it successfully completed, attributed to its greedy nature of minimizing the distance at any given moment.

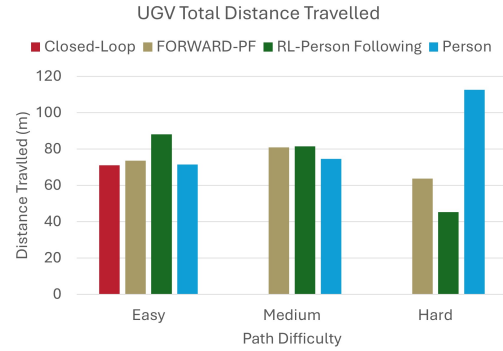


Fig. 7. Cumulative distance travelled by UGV. The performance of the methods are grouped by path difficulty. The blue illustrates the total distance travelled by the person. Where no bar is displayed, that method was unable to complete the experiment for the given path. The FORWARD-PF and RL-Person Following methods resulted in cumulative distances of the UGV less than the person's path.

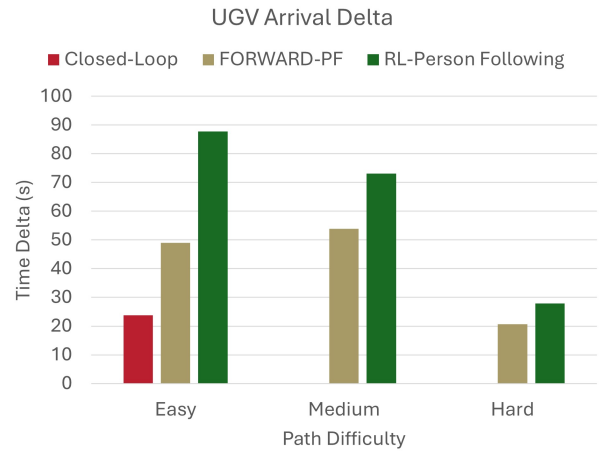


Fig. 8. Time delta between target reaching end position and UGV arriving at the targets end position. The performance of the methods are grouped by path difficulty. Where no bar is displayed, that method was unable to complete the experiment for the given path.

Observations of the robot behaviors revealed that the PF and RL methods averaged higher Time in View percentages, indicating better target visibility and up-to-date target location knowledge. The lower Max Distance for the PF method suggested more efficient decision-making compared to the RL method, although the PF method likely required fewer path replans. Notably, emergent behaviours like taking shortcuts were observed, especially with the PF method, enhancing path efficiency. Future work might focus on understanding the target visibility threshold necessary for successful missions in aerial-ground robot teams.

An observed advantage of the aerial-ground robot team is that the slow-moving UGV can take shortcuts when appropriate. Without prior knowledge of the target's intentions or path, it is impossible to predict the shortest route to the target's future position. However, as the UGV is continuously

updated with the target's position by the UAV, it can plan the most efficient path to the current target location without needing to maintain line-of-sight. This behavior is illustrated in Fig. 9 and Fig. 10. Both the FORWARD-PF and RL methods exhibited this behavior during experiments for path 3, and the FORWARD-PF method showed this behavior again for path 9. In a real-world validation experiment shown in Fig. 5, we were able to replicate the shortcut behavior particularly when implementing the FORWARD-PF method on a Jackal UGV and Crazyflie UAV. This simple experiment confirmed the possibility of such emergent behavior occurring in real-world implementations.

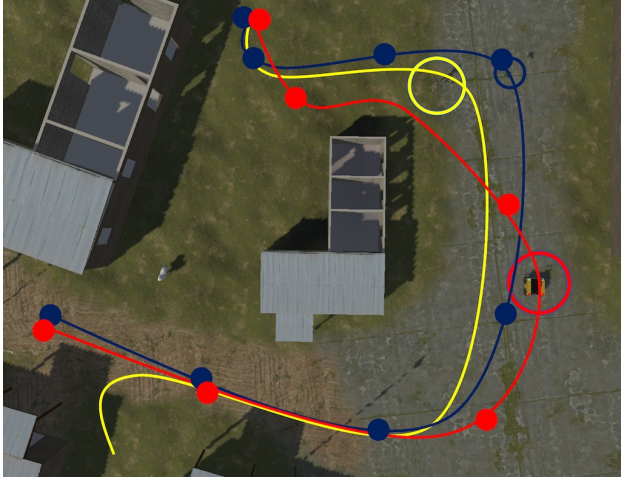


Fig. 9. Target (Yellow), UAV (Blue), UGV (Red). Paths showing method that follows the path of the target either through design or the UGV having equal velocity to the target which would naturally lead it to take a similar path.

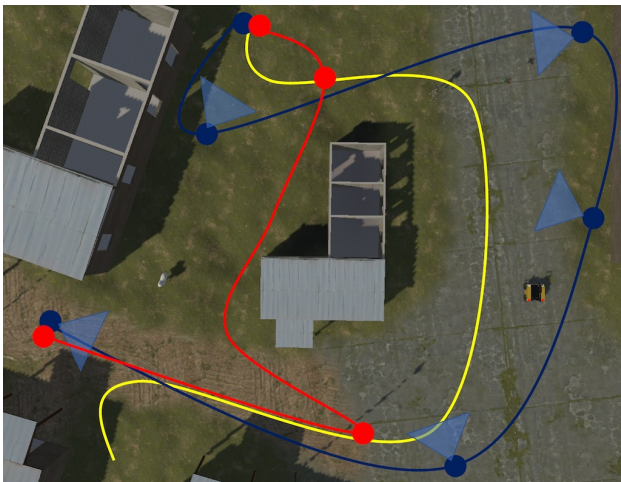


Fig. 10. Target (Yellow), UAV (Blue), UGV (Red). Paths showing the UGV taking a shortcut to navigate to the target with a decreased path length. The UAV is seen still following a similar path to the target as to maintain visibility.

VII. CONCLUSION

In this work, we present a multi-robot person-following problem subject to altitude, robot speed, and sensing con-

straints. We introduce two new methods: FORWARD-PF, a heuristic person-following method, and a MARL-based method for addressing this problem's challenges, and experimentally compare their performance against a closed-loop baseline method. In the course of accomplishing this, we also addressed challenges in adapting models trained in 2D to more complex 3D environments. Our results show that differences in sensor data, environmental dynamics, and agent interactions create barriers that affect the model's performance, highlighting the need for robust transfer methods. A notable limitation is the time and effort disparity between developing algorithms like FORWARD-PF and training MARL models. Although FORWARD-PF shows superior performance in simulations, the value lies in deploying heterogeneous systems for complex tasks, emphasizing that method selection is contingent on specific use cases.

Future work should establish thresholds for target visibility percentages in heterogeneous robot operations to optimize system reliability. Real hardware evaluations are necessary, requiring advanced aerial platforms to extend experimental capabilities. Investigating domain adaptation and its implications on model deployment across platforms will further inform performance constraints. This research primarily aimed to assess the robustness and efficiency of person-following methods. While social acceptance of these technologies remains unexplored here, it is also crucial as human-robot interactions increase.

REFERENCES

- [1] M. Natarajan, E. Seraj, B. Altundas, R. Paleja, S. Ye, L. Chen, R. Jensen, K. C. Chang, and M. Gombolay, "Human-Robot Teaming: Grand Challenges," *Current Robotics Reports*, vol. 4, no. 3, pp. 81–100, Aug. 2023.
- [2] S. S. O V, R. Parasuraman, and R. Pidaparti, "Impact of Heterogeneity in Multi-Robot Systems on Collective Behaviors Studied Using a Search and Rescue Problem," in *2020 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, Nov. 2020, pp. 290–297.
- [3] "Clearpath robotics warthog," <https://clearpathrobotics.com/warthog-unmanned-ground-vehicle-robot/>.
- [4] M. J. Islam, J. Hong, and J. Sattar, "Person-following by autonomous robots: A categorical overview," *The International Journal of Robotics Research*, vol. 38, no. 14, pp. 1581–1618, Dec. 2019.
- [5] L. J. Guibas, J.-C. Latombe, S. M. Lavalley, D. Lin, and R. Motwani, "A visibility-based pursuit-evasion problem," *International Journal of Computational Geometry & Applications*, vol. 09, no. 04n05, pp. 471–493, Aug. 1999.
- [6] L. Yuan, C. Reardon, G. Warnell, and G. Loianno, "Human Gaze-Driven Spatial Tasking of an Autonomous MAV," *IEEE Robotics and Automation Letters*, vol. 4, no. 2, pp. 1343–1350, Apr. 2019.
- [7] C. Reardon and J. Fink, "Air-ground robot team surveillance of complex 3D environments," in *2016 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*. Lausanne, Switzerland: IEEE, Oct. 2016, pp. 320–327.
- [8] M. Kouzeghar, Y. Song, M. Meghjani, and R. Bouffanais, "Multi-Target Pursuit by a Decentralized Heterogeneous UAV Swarm using Deep Multi-Agent Reinforcement Learning," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. London, United Kingdom: IEEE, May 2023, pp. 3289–3295.
- [9] R. Garcia, R. Strudel, S. Chen, E. Arlaud, I. Laptev, and C. Schmid, "Robust Visual Sim-to-Real Transfer for Robotic Manipulation," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2023, pp. 992–999.
- [10] A. Ghimire, X. Zhang, S. Javed, J. Dias, and N. Werghi, "Robot Person Following in Uniform Crowd Environment," May 2022.